EXAMINATIONS COUNCIL OF ZAMBIA

Examination for General Certificate of Education Ordinary Level

4030-2-000339

Additional Mathematics 4030/2

Paper 2

2021

Additional Materials:
Answer Booklet
Silent electronic calculator (non programmable)

Time: 2 hours 30 Minutes **Marks: 100**

Instructions to Candidates

1 Write the **centre number** and your **examination number** on **every page** of the separate **answer booklet** provided.

2 There are **twelve** questions in this paper. Answer **all** questions.

3 Write your answers in the separate Answer Booklet provided.

4 If you use more than one Answer Booklet, fasten the Answer Booklets together.

5 Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

Information for Candidates

1 The number of marks is shown in brackets [ ] at the end of each question or part question.

2 **The use of a non programmable electronic calculator is expected, where appropriate.**

3 You are reminded of the need for clear presentation in your answers.

4 Cell phones and other electronic devices are **not allowed** in the examination room.

5 Check the formulae overleaf.

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This question paper consists of 5 printed pages

Mathematical Formulae

1 ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

2 SERIES

Arithmetic $S_n = \frac{1}{2} n [2a + (n-1) d]$

Geometric $S_n = \frac{a(1-r^n)}{1-r}$ $(r \neq 1)$

$$S\infty = \frac{a}{1-r} \text{ for } |r| < 1$$

3 TRIGONOMETRY

Identities

$\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B.$

$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B.$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$\sin 2A = 2 \sin A \cos B$

$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

4 STATISTICS

Mean and standard deviation

Ungrouped data

$$\text{Mean } (\bar{x}) = \frac{\Sigma x}{n}, \text{ SD} = \sqrt{\left\{\frac{\Sigma (x-\bar{x})^2}{n}\right\}} = \sqrt{\left\{\frac{\Sigma x^2}{n} - (\bar{x})^2\right\}}$$

Grouped data

$$\text{Mean } (\bar{x}) = \frac{\Sigma fx}{\Sigma f}, \text{ SD} = \sqrt{\left\{\frac{\Sigma f(x-\bar{x})^2}{\Sigma f}\right\}} = \sqrt{\left\{\frac{\Sigma fx^2}{\Sigma f} - (\bar{x})^2\right\}}$$

1 Solve the following systems of equations

$2x + 3y + z = 17$,

$x - 3y + 2z = -8$,

$5x - 2y + 3z = 5$. [6]

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2 **(a)** Find the range of values of u for which $3u^2 - 8u < 3$. [3]

(b) Express $-3x^2 + 24x - 38$ in the form $a(x + b)^2 + c$, where a, b and c are constants. Hence, find the coordinates of the turning point. [4]

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3 Solve the equations

(a) $13^{3x-4} = \left(\frac{1}{4}\right)^{-3}$, [3]

(b) $\log_3(x + 25) = 2 + \log_3(2x - 1)$. [4]

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4 **(a)** Find the value of q, given that the expression $x^3 + 3x^2 + qx - 24$ is divisible by $(x - 3)$. [3]

(b) Solve the equation $x^3 - 7x + 6 = 0$. [4]

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5 **(a)** A group of 5 children consists of 2 girls whose names are Mary and Sarah, and 3 boys whose names are John, Joseph and James.
If these children sit in a straight line on 5 chairs, find the number of different possible seating arrangements if

(i) Joseph must sit in the middle, [2]

(ii) Mary and John must sit together. [3]

(b) A group of 6 designers is to be chosen from 8 men and 5 women. Find the number of ways of choosing at least 5 men. [3]

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6 **(a)** Solve the equation $3\sin(60° - x) = \sin x$ for values of x in the range $0° \leq x \leq 360°$. [4]

(b) **(i)** Express $4\cos x - \frac{5}{3}\sin x$ in the form $R\cos(x + \alpha)$, where $R > 0$ and $0° < \alpha < 90°$. [3]

(ii) Hence, find the maximum value of $12\cos x - 5\sin x$. [1]

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7 **(a)** In an arithmetic progression, the sum of the first 5 terms is 50 and the sum of the next 10 terms is 325. Find

(i) the first term and the common difference, [3]

(ii) the sum of the first 13 terms of the progression. [1]

(b) A doctor earns an annual salary of K4 000.00 during the first year of practice. In each succeeding year, the annual salary increases by 10%.

(i) Find the sum of the doctor's annual salaries over the first 6 years. [2]

(ii) Find the number of years the doctor must work, if the sum of his annual salaries is to exceed a million kwacha. [3]

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8 The table below shows the heights of 80 plants in a particular forest.

Height (cm)	164 - 168	169 - 173	174 - 178	179 - 183	184 - 188
Frequency	8	18	28	21	5

(a) Find the median class. [1]

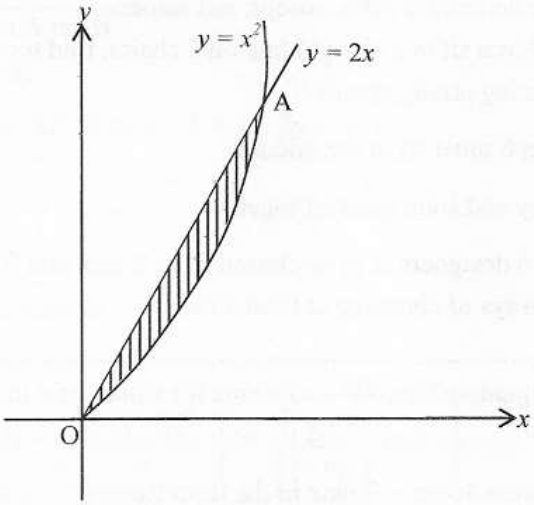
(b) Calculate

(i) an estimate of the mean, [2]

(ii) the standard deviation. [6]

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9 The following diagram shows part of the curve $y = x^2$ and the line $y = 2x$ meeting at A.



Find

(a) the coordinates of A, [3]

(b) the volume obtained by rotating the shaded region through 360° about the x-axis. [6]

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10 The velocity, Vm/s, of a particle moving in a straight line, at time t seconds is given by $V = 3t^2 - 2t + 2$. When $t = 0$, the particle is 3m from a fixed point O on the line.

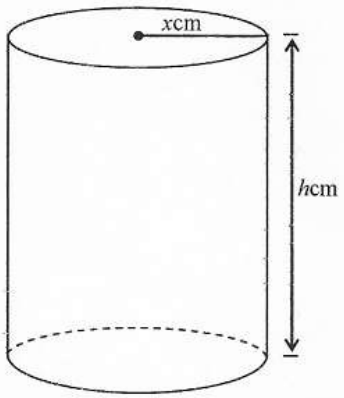
(a) Find the acceleration when $t = 1.5$. [2]

(b) Find an expression in terms of t, for its distance from the fixed point. [4]

(c) Find the distance between $t = 1$ and $t = 2$. [4]

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11 The following diagram shows a cylindrical metal plate of radius $x$cm and height, $h$cm.



(a) Given that its volume is 500cm^3, express h in terms of x. [2]

(b) Hence show that the total surface area (A) of the cylinder is given by

$$A = 2\pi x^2 + \frac{1000}{x}\text{cm}^2.$$ [3]

(c) Given that x varies, find the stationary value of A and show that this value is a minimum. [5]

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12 The curve $y = e^{-2x} - 3$ meets the x-axis at P and the y-axis at Q.

(a) Find the coordinates of P and of Q. [3]

(b) Sketch the graph of $y = e^{-2x} - 3$ for the domain $-1 \le x \le 1$. [3]

(c) Find the equation of the straight line which must be drawn on the graph of $y = e^{-2x} - 3$ to obtain a solution of the equation $x = \ln\left(\frac{1}{\sqrt{x+5}}\right)$. [4]